

Table 2. Incipience of boiling from enhanced surfaces is always associated with the trapping of either vapor or noncondensable gas on the surface. The bubble embryo radius for each incipience temperature condition was calculated based on the standard application of the Young-Laplace and Clausius-Clapeyron equations [30] in order to demonstrate the trapping effectiveness of the surface. Incipience temperatures were always lower for water than for HFE-7300. Other observations include:

- (1) the application of carbon nanotubes to a flat surface reduces boiling incipience temperatures, significantly reducing the temperature excursion common for a highly wetting liquid;
- (2) the comparatively large-scale sintered coating is more effective than carbon nanotubes at reducing the incipience temperature;
- (3) the hybrid surface virtually eliminated the incipience temperature for the highly wetting HFE-7300; however, in water there was no apparent incipience benefit for the hybrid